

BDA Module – MERN Stack Internship Project

This is a beginner-friendly BDA (Business Development Associate) Management System project.

Features:

- Add Leads
- View Leads
- Update Lead Status
- Delete Leads
- Dashboard Cards
- Responsive UI

1. FRONTEND (HTML + CSS + JS VERSION)

Create 3 files:

- index.html
- style.css
- script.js

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>BDA Dashboard</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="container">

    <h1>BDA Management System</h1>

    <div class="cards">
      <div class="card">
        <h2 id="totalLeads">0</h2>
```

```

    <p>Total Leads</p>
</div>

<div class="card">
    <h2 id="interestedLeads">0</h2>
    <p>Interested</p>
</div>

<div class="card">
    <h2 id="closedLeads">0</h2>
    <p>Closed</p>
</div>
</div>

<div class="form-section">
    <h2>Add New Lead</h2>

    <input type="text" id="name" placeholder="Client Name">
    <input type="text" id="company" placeholder="Company Name">
    <input type="email" id="email" placeholder="Email">

    <select id="status">
        <option value="New">New</option>
        <option value="Interested">Interested</option>
        <option value="Closed">Closed</option>
    </select>

    <button onclick="addLead()">Add Lead</button>
</div>

<div class="table-section">
    <h2>Lead List</h2>

    <table>
        <thead>
            <tr>
                <th>Name</th>
                <th>Company</th>
                <th>Email</th>
                <th>Status</th>
                <th>Action</th>
            </tr>
        </thead>

        <tbody id="leadTableBody">

            </tbody>
    </table>

```

```
    </div>

</div>

<script src="script.js"></script>
</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  font-family: Arial, sans-serif;
}

body {
  background-color: #f4f7fc;
  padding: 30px;
}

.container {
  max-width: 1200px;
  margin: auto;
}

h1 {
  margin-bottom: 30px;
  text-align: center;
}

.cards {
  display: flex;
  gap: 20px;
  margin-bottom: 30px;
}

.card {
  flex: 1;
  background: white;
  padding: 25px;
  border-radius: 10px;
  box-shadow: 0 2px 10px rgba(0,0,0,0.1);
}
```

```

    text-align: center;
}

.form-section,
.table-section {
    background: white;
    padding: 25px;
    border-radius: 10px;
    margin-bottom: 30px;
    box-shadow: 0 2px 10px rgba(0,0,0,0.1);
}

.form-section input,
.form-section select {
    width: 100%;
    padding: 12px;
    margin-top: 15px;
    border: 1px solid #ccc;
    border-radius: 6px;
}

button {
    margin-top: 15px;
    padding: 12px 20px;
    background: #007bff;
    color: white;
    border: none;
    border-radius: 6px;
    cursor: pointer;
}

button:hover {
    background: #0056b3;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 20px;
}

th,
td {
    border: 1px solid #ddd;
    padding: 12px;
    text-align: center;
}

```

```
th {
  background-color: #007bff;
  color: white;
}

.delete-btn {
  background-color: red;
}
```

script.js

```
let leads = [];

function addLead() {

  const name = document.getElementById("name").value;
  const company = document.getElementById("company").value;
  const email = document.getElementById("email").value;
  const status = document.getElementById("status").value;

  if(name === "" || company === "" || email === "") {
    alert("Please fill all fields");
    return;
  }

  const lead = {
    id: Date.now(),
    name,
    company,
    email,
    status
  };

  leads.push(lead);

  displayLeads();
  updateDashboard();

  document.getElementById("name").value = "";
  document.getElementById("company").value = "";
  document.getElementById("email").value = "";
}

function displayLeads() {
```

```

const tableBody = document.getElementById("leadTableBody");

tableBody.innerHTML = "";

leads.forEach((lead) => {

  tableBody.innerHTML += `
    <tr>
      <td>${lead.name}</td>
      <td>${lead.company}</td>
      <td>${lead.email}</td>
      <td>${lead.status}</td>
      <td>
        <button class="delete-btn" onclick="deleteLead(${lead.id})">
          Delete
        </button>
      </td>
    </tr>
  `;
});
}

function deleteLead(id) {

  leads = leads.filter((lead) => lead.id !== id);

  displayLeads();
  updateDashboard();
}

function updateDashboard() {

  document.getElementById("totalLeads").innerText = leads.length;

  const interested = leads.filter(
    (lead) => lead.status === "Interested"
  ).length;

  const closed = leads.filter(
    (lead) => lead.status === "Closed"
  ).length;

  document.getElementById("interestedLeads").innerText = interested;
  document.getElementById("closedLeads").innerText = closed;
}

```

HOW TO RUN FRONTEND VERSION

1. Create a folder
2. Add the 3 files
3. Open index.html in browser

2. MERN STACK VERSION

Folder Structure:

```
bda-project/  
├── backend/  
│   ├── server.js  
│   ├── models/  
│   │   └── Lead.js  
│   └── routes/  
│       └── leadRoutes.js  
└── frontend/  
    └── React App
```

BACKEND SETUP

Step 1

Create backend folder.

Run:

```
npm init -y
```

Install packages:

```
npm install express mongoose cors nodemon
```

server.js

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();

app.use(cors());
app.use(express.json());

mongoose.connect("mongodb://127.0.0.1:27017/bdaDB")
  .then(() => console.log("MongoDB Connected"))
  .catch((err) => console.log(err));

const leadRoutes = require("./routes/leadRoutes");

app.use("/api/leads", leadRoutes);

app.listen(5000, () => {
  console.log("Server running on port 5000");
});
```

models/Lead.js

```
const mongoose = require("mongoose");

const leadSchema = new mongoose.Schema({
  name: String,
  company: String,
  email: String,
  status: String
});

module.exports = mongoose.model("Lead", leadSchema);
```


routes/leadRoutes.js

```
const express = require("express");
const router = express.Router();

const Lead = require("../models/Lead");

router.post("/", async (req, res) => {

  const lead = new Lead(req.body);

  await lead.save();

  res.json(lead);
});

router.get("/", async (req, res) => {

  const leads = await Lead.find();

  res.json(leads);
});

router.delete("/:id", async (req, res) => {

  await Lead.findByIdAndDelete(req.params.id);

  res.json({ message: "Lead Deleted" });
});

module.exports = router;
```

FRONTEND REACT SETUP

Step 1

Create frontend:

```
npx create-react-app frontend
```

Install axios:

```
npm install axios
```

src/App.js

```
import { useEffect, useState } from "react";
import axios from "axios";
import "./App.css";

function App() {

  const [leads, setLeads] = useState([]);

  const [formData, setFormData] = useState({
    name: "",
    company: "",
    email: "",
    status: "New"
  });

  useEffect(() => {
    fetchLeads();
  }, []);

  const fetchLeads = async () => {

    const res = await axios.get("http://localhost:5000/api/leads");

    setLeads(res.data);
  };

  const handleChange = (e) => {

    setFormData({
      ...formData,
      [e.target.name]: e.target.value
    });
  };

  const addLead = async () => {

    await axios.post(
      "http://localhost:5000/api/leads",
      formData
    );
  };
}
```

```

    );

    fetchLeads();
  };

  const deleteLead = async (id) => {

    await axios.delete(
      `http://localhost:5000/api/leads/${id}`
    );

    fetchLeads();
  };

  return (
    <div className="container">

      <h1>BDA CRM Dashboard</h1>

      <div className="form">

        <input
          type="text"
          placeholder="Name"
          name="name"
          onChange={handleChange}
        />

        <input
          type="text"
          placeholder="Company"
          name="company"
          onChange={handleChange}
        />

        <input
          type="email"
          placeholder="Email"
          name="email"
          onChange={handleChange}
        />

        <select name="status" onChange={handleChange}>
          <option>New</option>
          <option>Interested</option>
          <option>Closed</option>
        </select>

```

```

        <button onClick={addLead}>Add Lead</button>
      </div>

      <div className="lead-list">

        {
          leads.map((lead) => (

            <div className="lead-card" key={lead._id}>

              <h3>{lead.name}</h3>

              <p>{lead.company}</p>

              <p>{lead.email}</p>

              <p>{lead.status}</p>

              <button onClick={() => deleteLead(lead._id)}>
                Delete
              </button>

            </div>
          ))
        }

      </div>

    </div>
  );
}

export default App;

```

src/App.css

```

body {
  background: #f4f7fc;
  font-family: Arial;
}

.container {
  max-width: 1000px;
  margin: auto;
}

```

```
padding: 20px;
}

.form {
  display: flex;
  flex-direction: column;
  gap: 10px;
  background: white;
  padding: 20px;
  border-radius: 10px;
}

.form input,
.form select {
  padding: 10px;
}

button {
  padding: 10px;
  cursor: pointer;
}

.lead-list {
  margin-top: 20px;
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 20px;
}

.lead-card {
  background: white;
  padding: 20px;
  border-radius: 10px;
}
```

HOW TO RUN MERN PROJECT

Start MongoDB

Make sure MongoDB is installed.

Run Backend

```
cd backend  
nodemon server.js
```

Run Frontend

```
cd frontend  
npm start
```

EXTRA FEATURES YOU CAN ADD

To impress the company more:

- Search Leads
 - Filters
 - JWT Authentication
 - Dark Mode
 - Charts
 - Pagination
 - Team Member Assignment
 - Lead Notes
 - Export to Excel
 - Responsive Sidebar
-

WHAT TO SAY IN INTERVIEW

"I built a BDA management dashboard using MERN stack where users can create, manage, track, and delete client leads. The project includes CRUD operations, REST APIs, MongoDB integration, and a responsive frontend dashboard."